

Topic:

Managing Privileges

Privileges:

- ⇒ Permitting only certain users to access, process and alter data.
- ⇒ Privilege is a right to execute a particular type of SQL statement or to access another user's object.

Types of Privileges

⇒ There are two types of privileges:

1. System privileges:

- o General commands privileges.
- o Admin tasks
- o Should be restricted
- o Not everyone granted.

⇒ Given to users to allow them to perform certain functions that deal with managing the database and the users.

⇒ Privileges that control access to actions in the database are called system privileges.

⇒ Grant the right to perform a specific type of action in the database.

⇒ There are more than 100 distinct system privileges.

⇒ Includes:

- CREATE Table
- CREATE SESSION
- CREATE USER
- CREATE TABLESPACE
- DROP Table
- altering tablespaces
- Creating users.

Category	Examples
INDEX	CREATE ANY INDEX
	ALTER ANY INDEX
	DROP ANY INDEX
TABLE	CREATE TABLE
	CREATE ANY TABLE
	ALTER ANY TABLE
	DROP ANY TABLE
	SELECT ANY TABLE
	UPDATE ANY TABLE
SESSION	DELETE ANY TABLE
	CREATE SESSION
	ALTER SESSION
TABLESPACE	RESTRICTED SESSION
	CREATE TABLESPACE
	ALTER TABLESPACE
	DROP TABLESPACE
	UNLIMITED TABLESPACE

Note:

ANY: Signifies that users have the privileges in any schema.

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GRANT:

- Adds a privilege to a user or a group of users.

REVOKE:

- Deletes the privileges.

Granting System Privileges

- ⇒ Use the GRANT keywords to grant system privileges.
- ⇒ Grantee can further grant the system privilege with ADMIN option.

Syntax:

- ⇒ GRANT <system-privilege> TO <grantee clause> [WITH ADMIN

EX:

- ⇒ GRANT CREATE SESSION TO emi WITH ADMIN OPTION

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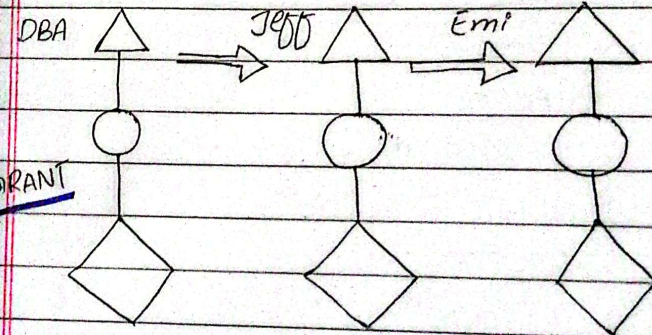
Revoking System Privileges

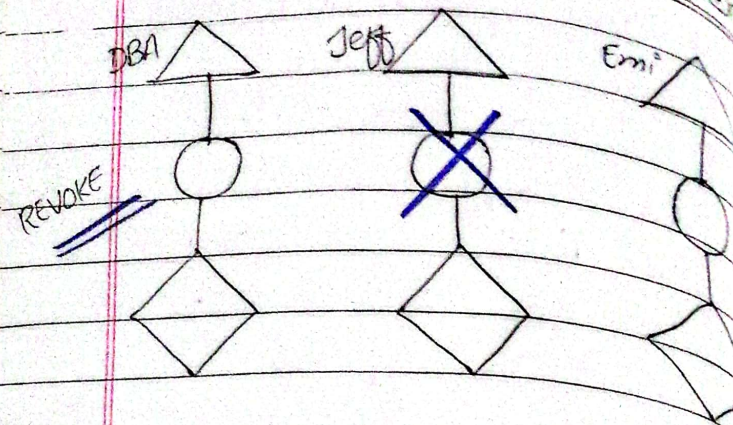
- Use the REVOKE command to remove a system privilege from a user.
- Users with ADMIN OPTION for system privilege can revoke system privileges.
- Can only revoke privileges with a GRANT command.

Syntax / EX:

- REVOKE CREATE TABLE emp;

Revoking system privileges with the admin options





2. Object privileges:

⇒ Use to perform a particular action on specific objects such as table, view, sequence, procedure, function or package.

⇒ Includes:

- SELECT
- UPDATE
- ALTER
- DELETE
- EXECUTE
- INDEX
- INSERT
- REFERENCES

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○ These are the privileges that control access to data and other objects are called object privileges.

○ Object privileges are granted on a specific object in the database.

Syntax:

GRANT <object privilege> ON
<object> TO <grantee clause>
[WITH GRANT OPTION].

Granting object Privileges

⇒ Use the GRANT command to grant object privilege.

⇒ Grant must be in grantor's schema or grantor must have GRANT OPTION.

Ex:

GRANT UPDATE ON emi.customers
TO Jeff WITH GRANT OPTION;

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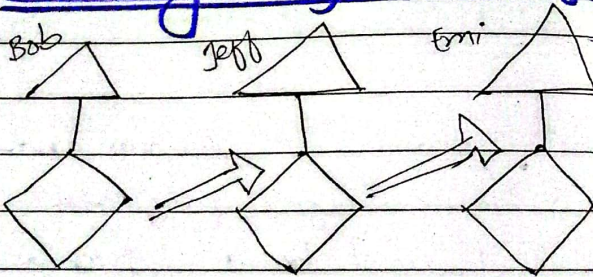
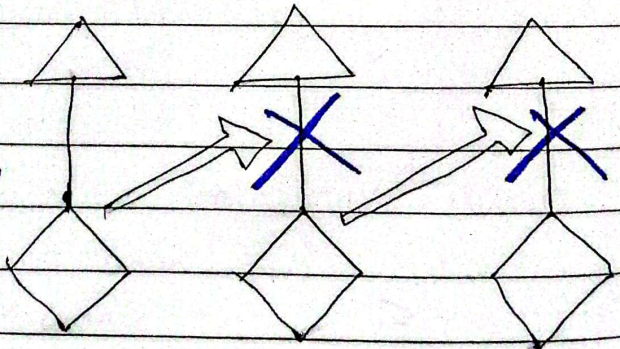
Revoking object privileges

- Use the REVOKE command to revoke object privileges.
- Uses revoking the privilege must be the original grantor of the object privilege being revoked.

Ex:

◦ REVOKE SELECT ON emp.Oracle
FROM jeff;

Revoking object Privilege

GRANTREVOKE

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Obtaining privileges information

⇒ Information about privileges can be obtained by querying the following views:

- DBA_SYS_PRIVS
- SESSION_PRIVS
- DBA_TAB_PRIVS
- DBA_COL_PRIVS